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Analytical Ability

Instructions [1 - 20]

In the following questions a question is followed by data in the form of two statements labelled as I and II. You must decide whether the data given in the statements are sufficient to answer the questions. Using the data make an appropriate choice from (a) to (d) as per the following guidelines :

1. What is the 100^{th} term of a sequence ?

I. The sequence is an Arithmetic Progression.

II. The first term of the sequence is 5.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

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2. Can we find the quadratic equation $ax^2 + bx + c = 0$?

I. The sum of the roots is given.

II. The difference of the roots is given.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

3. Do the two circles with their centres at A and B and having radii r_1 , and r_2 , intersect ?

I. Distance between A and B is 'a' units.

II. $|r_1 - r_2| < a < r_1 + r_2$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.

- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

4. **Are two triangles similar ?**

- I. They are equilateral.**
II. Their areas are equal.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

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5. **What is the area of a regular hexagon ?**

- I. A circle of radius 'a' units is inscribed in the hexagon.**
II. Each side of the hexagon subtends an angle of 60° at the centre of the circle.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

6. **What is the value of $\log_{10} x$?**

- I. $x > 3$**
II. $\log_{10} x + \log_{10}(x - 3) = 1$

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.

- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

7. Is ABCD a cyclic quadrilateral ?

I. The lengths of the diagonals are given.

II. A circle with mid point of a diagonal as centre passes through the vertices.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

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8. What is the angle subtended by the chord AB of a circle at a point P on the circumference ?

I. The radius of the circle is 5 cm.

II. The length of the chord AB is 5 cm.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

9. What is the value of $119x + 247y$?

I. $118x + 246y = 482$

II. x is greater than y by 2

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.

- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

10. **What is the percentage of profit on the sale of 50 books ?**

I. The cost price of each book is ₹ 100.

II. The sale price of each book is ₹ 125.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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11. **What is the value of n ?**

I. $1^3 + 2^3 + 3^3 + \dots + n^3 = 225$

II. n is positive integer

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

12. **What is the day of 31st December of an year ?**

I. In that year the first of March was Monday.

II. That year was a leap year.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

13. What is the Arithmetic Mean of the numbers x , y , z and t ?

I. $x + 2y - z - 3t = K$

II. $2x + y + 4z + 6t = L$

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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14. What is the equation of a straight line ?

I. The line is parallel to $2x + 3y = 5$.

II. The line does not pass through the origin.

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

15. What is the value of $8a^3 + b^3 - 27c^3$?

I. $abc = k$

II. $2a + b = 3c$

- A** if the statement I alone is sufficient to answer the question.
- B** if the statement II alone is sufficient to answer the question.
- C** if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D** if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

16. What is the distance between two ships ?

I. Seen from the top of a mountain their angles of depression are 60° and 30° .

II. Height of the mountain is 60 metres.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

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17. Is $(x - a)$ a factor of the polynomial $f(x)$?

I. $f(a) \neq 0$

II. The sum of the coefficients of $f(x)$ is constant.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

18. What is the simple interest earned yearly on a deposit in a bank ?

I. The amount deposited is ₹ 10,000.

II. The rate of interest is 8%.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

19. What is the surface area of a sphere ?

I. The sphere is made of iron.

II. The radius of the sphere is given.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

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20. What is the value of $x^4 + y^4$, if $xy = 1$?

I. $x + y = 5$

II. $x > 0, y > 0$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

Instructions [21 - 30]

In the following Questions a sequence of numbers and letters that follow a definite pattern is given. Each question has a blank space. This blank space is to be filled by the correct answer from one of the four given options to complete the sequence without breaking the pattern.

21. 122 : 145 :: 257 :

- A 290
- B 299
- C 401
- D 497

Answer: A

22. 1001 : 1332 :: : 2198

A 1728

B 1729

C 2413

D 2526

Answer: B

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23. **ABCD : ZYXW :: : VTUS**

A EFHG

B EHGF

C EGFH

D EFGH

Answer: C

24. **Tiger : Goat :: : Herbivore**

A Carivore

B Omnivore

C Parasite

D Insectivore

Answer: A

25. **Quadrilateral : Square :: Triangle :**

A Equilateral triangle

B Isosceles triangle

C Scalene triangle

D Right-angled triangle

Answer: A

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26. **Wood : Chair :: : Ornament**

- A** Coal
- B** Gold
- C** Hammer
- D** Bauxite

Answer: B

27. Ice : Water :: : Ghee

- A** Milk
- B** Curd
- C** Butter
- D** Oil

Answer: C

28. $\frac{n(n+1)}{2} : \sum n :: : \sum n^2$

- A** $\frac{n^2(n-1)^2}{4}$
- B** $\frac{n^2(n+1)^2}{4}$
- C** $\frac{n(n-1)(2n-1)}{6}$
- D** $\frac{n(n+1)(2n+1)}{6}$

Answer: D

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29. ABCDEFG : STUVXYZ :: GECA :

- A** ZXUS
- B** ZXSU
- C** ZSXU
- D** ZSUX

Answer: A

30. Volume of a cylinder : Area of curved surface of the cylinder :: $\pi r^2 h$:

- A** πrh
- B** $2\pi rh$
- C** $\frac{2}{3}\pi rh$
- D** $\frac{1}{2}\pi r^2 h$

Answer: B

Instructions [31 - 35]

In the following questions pick the odd thing out :

- A** $\sin^2 x + \cos^2 x$
- B** $\sec^2 x - \tan^2 x$
- C** $\tan^2 x + \cot^2 x$
- D** $\operatorname{cosec}^2 x - \cot^2 x$

Answer: C

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- A** Eagle
- B** Vulture
- C** Owl
- D** Crow

Answer: C

- A** ABYZ
- B** CDWX
- C** EFUV
- D** GHTS

Answer: D

- A** $x^3 - 4 = 0$
- B** $x^3 - 3 = 0$

C $x^3 - 2 = 0$

D $x^3 - 1 = 0$

Answer: D

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A (5, 6, 11)

B (6, 7, 13)

C (7, 8, 15)

D (8, 9, 19)

Answer: D

Instructions [31 - 40]

Each of the questions follows definite pattern. Observe the same and fill in the blanks with suitable answers.

31. $\begin{matrix} 1 & 5 & 10 & 17 & & 37 \\ 5, & 10, & 17, & 26, & \dots\dots\dots, & 50 \end{matrix}$

A $\begin{matrix} 26 \\ 37 \end{matrix}$

B $\begin{matrix} 36 \\ 37 \end{matrix}$

C $\begin{matrix} 26 \\ 50 \end{matrix}$

D $\begin{matrix} 36 \\ 49 \end{matrix}$

Answer: A

32. 6, 12, 20, 30,, 56

A 50

B 48

C 44

D 42

Answer: D

Explanation:

The logic here is

$$12 - 6 = 6$$

$$20 - 12 = 8$$

$$30 - 20 = 10$$

The difference between the consecutive numbers is increasing by 2.

The next difference should be 12.

∴ The missing number is $30+12 = 42$

Hence, the correct answer is Option D

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33. $\frac{1}{2}, \frac{8}{5}, \frac{27}{10}, \dots, \frac{125}{26}, \frac{37}{50}$

A $\frac{36}{17}$

B $\frac{32}{13}$

C $\frac{64}{17}$

D $\frac{81}{25}$

Answer: C

Explanation:

Numerator

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

Denominator

$$5-2 = 3$$

$$10-5 = 5$$

next difference should be 7

$$\text{Hence } 10 + 7 = 17$$

34. 4, 1, 9, 3, 16, 6, 25, 10, 36,

A 15

B 20

C 25

D 30

Answer: A

35. 1, 4, 27, 256,

A 625

B 3152

C 1024

D 3125

Answer: D

Explanation:

The logic here is

$$1^1 = 1$$

$$2^2 = 4$$

$$3^3 = 27$$

$$4^4 = 256$$

Similarly, the next number in the series = $5^5 = 3125$

Hence, the correct answer is Option D

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36. $1, 2, 6, 24, \dots, 720, 5040$

A $\frac{1}{100}$

B $\frac{1}{120}$

C $\frac{1}{72}$

D $\frac{1}{240}$

Answer: B

37. 4, 7, 12, 19, 28,, 52

A 39

B 40

C 41

D 42

Answer: A

Explanation:

The pattern followed is: the difference between the numbers is increasing by 2 starting from 3.

$$\text{i.e } 4+3 = 7$$

$$7+5 = 12$$

$$12+7 = 19$$

$19 + 9 = 28$

$28 + 11 = 39$

38. 1 x 8, 3 v 6, 5 t 4,

A 7 r 3

B 7 r 2

C 7 v 2

D 7 v 3

Answer: B

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39. 1, 2, 3, 5, 8, 13, 21,

A 24

B 26

C 29

D 34

Answer: D

Explanation:

The logic here is

$1 + 2 = 3$

$2 + 3 = 5$

$3 + 5 = 8$

$5 + 8 = 13$

$8 + 13 = 21$

From 3rd number, every number is the sum of the previous two numbers.

Hence the next number would be $21 + 13 = 34$.

40. 5, 6, 10, 19, 35, 60,

A 85

B 92

C 96

D 105

Answer: C

Instructions [41 - 43]

The following table gives the details of the 5 commodities A, B, C, D, E required, with their costs, for a family in a month.

Study the table and answer the questions.

Commodity	Quantity required/month(in	Rate/kg in ₹	
		Year 1990	Year 1995
A	10	70	100
B	15	40	45
C	8	35	40
D	25	50	60
E	5	58	62

41. The total amount(in ₹) spent on all the 5 commodities by the family in 1990 is

A 253

B 315

C 302

D 312

Answer: E

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42. The percentage of increase(per kg) in the rate of the commodity E from 1990 to 1995 is

A 20.3%

B 6.89%

C 345%

D 17.8%

Answer: B

Explanation:

From the table,

The percentage of increase(per kg) in the rate of the commodity E from 1990 to 1995 is = $\frac{(62-58)}{58} \times 100 = \frac{4}{58} \times 100 = 6.89\%$

Hence, the correct answer is Option B

43. The amount (in ₹) spent extra on commodities B and C in the year 1995 than in the year 1990 is

A 115

B 230

C 260

D 180

Answer: A

Explanation:

Money spent on B in 1990 = $15 \times 40 = 600$

Money spent on C in 1990 = $8 \times 35 = 280$

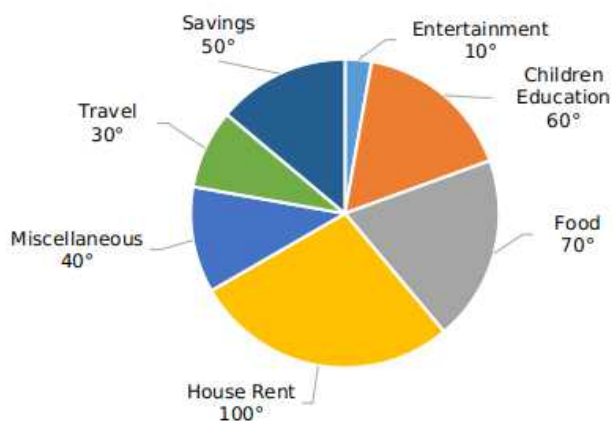
Money spent on B in 1995 = $15 \times 45 = 675$

Money spent on C in 1995 = $8 \times 40 = 320$

The amount (in ₹) spent extra on commodities B and C in the year 1995 than in the year 1990 = $(675 + 320) - (600 + 280) = 115$

Instructions [44 - 48]

The amounts spent by a person under various heads in a month is given in the following Pie Chart. Based on this information answer the following questions.



44. If the monthly income of the person is ₹ 54,000, then the difference between the amounts spent on travel and food per month (in ₹) is

A 21,600

B 16,200

C 6,000

D 12,000

Answer: C

Explanation:

Given, monthly income of the person is ₹ 54,000 = 360°

difference between the amounts spent on travel and food per month (in ₹) is = $70^\circ - 30^\circ = 40^\circ = 40 \cdot \frac{54000}{360} = 6000$

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45. If the amounts spent on savings and children education put together is ₹ 22,000, then the monthly income of the person (in ₹) is

- A 75,000
- B 66,000
- C 72,000
- D 36,000

Answer: C

Explanation:

$$\text{Savings} + \text{children's education} = 50^\circ + 60^\circ = 110^\circ$$

$$\text{Given, } 110^\circ = 22000$$

$$\text{monthly income} = 360^\circ = 360 \cdot \frac{22000}{110} = 72000$$

46. The amount spent on children education is equal to the amounts spent on which of the following two heads ?

- A Entertainment and miscellaneous
- B Travel and miscellaneous
- C Savings and travel
- D Savings and entertainment

Answer: D

Explanation:

$$\text{amount spent on children education} = 60$$

$$\text{A: Entertainment} + \text{misc} = 50$$

$$\text{B: Travel} + \text{mis} = 70$$

$$\text{C : savings} + \text{travel} = 80$$

$$\text{D : savings} + \text{enter} = 60$$

47. If the amount spent on house rent in a month is ₹ 5,000 more than the amount spent on travel and savings put together, then the monthly income (in ₹) of the person is

- A 25,000
- B 90,000
- C 1,00,000
- D 45,000

Answer: B

Explanation:

Let total amount = $360x$

amount spent on house rent = $100x$

amount spent on travel and savings put together = $80x$

Given, $100x - 80x = 5000$

$\Rightarrow x = 250$

Monthly income = $360 \times 250 = 90000$

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48. Monthly savings percentage is

A $13\frac{8}{9}\%$

B $13\frac{2}{9}\%$

C $13\frac{1}{9}\%$

D 14%

Answer: A

Explanation:

From the pie chart,

$$\text{Monthly savings percentage} = \frac{50^\circ}{360^\circ} \times 100 = \frac{125}{9} = 13\frac{8}{9}\%$$

Hence, the correct answer is Option A

Instructions [49 - 50]

In a group of 75 students 28 students study Mathematics (M), 32 study Physics (P), 30 study Chemistry (C) and 15 students study none of these; 13 study both M and P, 12 study P and C and 10 study C and M. Based on this information answer the questions.

49. The number of students who study only Chemistry is

A 13

B 14

C 15

D 16

Answer: A

Explanation:

Let a = only M

b = only P

c = only C

d = only M and P

e = only M and C

f = only P and C

g = all three

n = none = 15

$$a+b+c+d+e+f+g+n = 75 \Rightarrow a+b+c+d+e+f+g = 60$$

$$X = a+b+c$$

$$Y = d+e+f$$

$$Z = g$$

$$X+2Y + 3Z = 28 + 32 + 30 = 90 \text{ ----- (1)}$$

$$d+g = 13$$

$$e+g = 10$$

$$f+g = 12$$

$$\text{Adding we get } Y + 3Z = 35 \text{ -----(2)}$$

Subtracting 1 and 2, we get $X + Y = 55$

$$\Rightarrow Z = g = 5$$

Hence $d = 8$, $e = 5$ and $f = 7$ $a = 10$, $b = 12$, $c = 13$

Only chemistry = $c = 13$

50. The number of students who study Physics alone is

A 10

B 13

C 12

D 15

Answer: C

Explanation:

Let a = only M

b = only P

c = only C

d = only M and P

e = only M and C

f = only P and C

g = all three

n = none = 15

$$a+b+c+d+e+f+g+n = 75 \Rightarrow a+b+c+d+e+f+g = 60$$

$$X = a+b+c$$

$$Y = d+e+f$$

$$Z = g$$

$$X+2Y + 3Z = 28 + 32 + 30 = 90 \text{ ----- (1)}$$

$$d+g = 13$$

$$e+g = 10$$

$$f+g = 12$$

$$\text{Adding we get } Y + 3Z = 35 \text{ ----- (2)}$$

$$\text{Subtracting 1 and 2, we get } X + Y = 55$$

$$\Rightarrow Z = g = 5$$

$$\text{Hence } d= 8, e=5 \text{ and } f = 7 \text{ } a = 10, b = 12, c= 13$$

$$\text{Only physics } = b = 12$$

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Instructions [51 - 55]

The letters of English alphabet are arranged cyclically. Then the letters are coded as follows : (i) a vowel is to be coded as the second vowel after it in the clockwise direction. and (ii) a consonant is to be coded as the second consonant after it in the clockwise direction. For decoding, the inverse process is to be followed. Based on this coding and decoding system answer the questions.

51. **Code word for HAND is**

- A** KIQH
- B** KIQF
- C** KIPG
- D** KIQG

Answer: D

52. **The word that is coded as RIUT is**

- A** PAIR
- B** PAIW
- C** PAAR
- D** PUIR

Answer: A

53. **The word that is coded as RNIVO is**

- A** TLATE
- B** PLATA
- C** PLASE
- D** TQUXA

Answer: C

Explanation:

Based on the given conditions,

R will be the code for P

N will be the code for L

I will be the code for A

V will be the code for S

O will be the code for E

∴ The word that is coded as RNIVO is PLASE.

Hence, the correct answer is Option C

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54. The code word for LEAVE is

- A** NOIXO
- B** NOLXU
- C** JUOTO
- D** JUOTU

Answer: A

Explanation:

Based on the given conditions,

L will be coded as N

E will be coded as O

A will be coded as I

V will be coded as X

E will be coded as O

∴ The code word for LEAVE is NOIXO.

Hence, the correct answer is Option A

55. The word that is coded as QUEUE is

- A** NIOIO
- B** SEOEO
- C** NIUIU
- D** SIOIO

Answer: C

56. If the word **COMPUTER** is coded as **RFUVQNPC**, the code word for **MEDICINE** is

- A** EOJDJEFM
- B** EMJDJFEM
- C** EOKDKFEM
- D** EOLDLFEM

Answer: A

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57. If the word **DWUKPGUU** is coded as **BUSINESS**, the word **MANAGEMENT** is coded as

- A** KZLZECKCLR
- B** KXLXECKCLR
- C** KYLYECKCLR
- D** KYLYEDKDLR

Answer: C

58. If the word **CENTRAL** is coded as **LARTNEC** then the code word for **SEMINAR** is

- A** RINAMES
- B** RANMIES
- C** RAMNIES
- D** RANIMES

Answer: D

59. If the word **FLUTE** is coded as **EJRPZ**, then the code word of **PLANE** is

- A** OJXIZ

- B** OJXKZ
- C** OJXLZ
- D** OJXLA

Answer: A

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60. If the code word for VICTORY is XKEVQTA then the code word for SUCCESS is

- A** UWEEGUU
- B** VXFFHVV
- C** UWEEGVV
- D** UWEEHUU

Answer: A

61. If the first day of March in a year happens to be Friday, then the day on which Indian Independence day falls in that year is

- A** Monday
- B** Tuesday
- C** Thursday
- D** Saturday

Answer: C

62. If a clock shows 15 minutes past 3 O'clock then the angle between the hours and minutes hands is

- A** 20°
- B** 30°
- C** 40°
- D** 7.5°

Answer: D

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63. If the minutes hand of a clock is facing South, then the direction of the minutes hand after 210 minutes is

- A** North
- B** East
- C** West
- D** South-West

Answer: A

64. **A is the mother of B and C. D is the husband of C. Then A is related to D as**

- A** Mother-in-law
- B** Mother
- C** Daughter-in-law
- D** Father-in-law

Answer: A

Instructions [65 - 66]

Four buses B_1, B_2, B_3, B_4 , depart from the station S_1 , and arrive the stations S_2 , and S_3 . The bus numbers and the order in which they depart or arrive is not the same. The first bus to leave S_1 , is the second to reach S_3 , and the third to reach S_2 . The first bus to reach S_3 , is the second bus to leave S_1 , and the last bus to reach S_2 .

Based on this information answer the following questions.

65. **The first bus to reach the station S_1 is**

- A** B_1
- B** B_2
- C** B_3
- D** B_4

Answer: E

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66. **The second bus to reach the station S_2 is**

- A** B_1
- B** B_2
- C** B_3

D B_4

Answer: E

67. In a queue, Anitha is the 10^{th} from the front while Meena is 25^{th} from the last and Mohan is just in the middle of the two. If there are 50 persons in the queue, the position of Mohan from the front is

A 20^{th}

B 19^{th}

C 18^{th}

D 17^{th}

Answer: C

68. If $a * b = a + b + ab$ for $a, b \in Q$ then the value of x satisfying $(1 * 2) * x = -13$ is

A 1

B -3

C 2

D -2

Answer: B

Explanation:

Given, $a * b = a + b + ab$ and

$$(1 * 2) * x = -13$$

$$\Rightarrow (1 + 2 + 1.2) * x = -13$$

$$\Rightarrow 5 * x = -13$$

$$\Rightarrow 5 + x + 5x = -13$$

$$\Rightarrow 5 + 6x = -13$$

$$\Rightarrow 6x = -18$$

$$\Rightarrow x = -3$$

Hence, the correct answer is Option B

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69. $a * b = \frac{1}{ab} + 1 \Rightarrow \sum_{n=1}^{2013} n * (n + 1) =$

A $2013 - \frac{1}{2013}$

B $2014 - \frac{1}{2014}$

C $\frac{-1}{2014}$

D $2013 - \frac{1}{2014}$

Answer: B

70. If $a * b = a + b - \frac{ab}{2} \forall a, b \in R$ and if c is a non-zero real number, then the value of x satisfying $x * c = x$ is

A 1

B 2

C 3

D 5

Answer: B

Explanation:

Given, $a * b = a + b - \frac{ab}{2}$

If $x * c = x$

$\Rightarrow x + c - \frac{xc}{2} = x$

$\Rightarrow c - \frac{xc}{2} = 0$

$\Rightarrow \frac{xc}{2} = c$

$\Rightarrow x = 2$

$\therefore$ The value of x satisfying $x * c = x$ is $x = 2$

Hence, the correct answer is Option B

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Mathematical Ability

71. If $(36)^2 \cdot 9^4 \cdot (18)^2 = 2^\alpha \cdot 3^\beta$ then $\alpha + \beta =$

A -7

B -8

C 9

D 8

Answer: C

Explanation:

Solution

$$8^3 \cdot (27)^4 \cdot 6^5 = \left((2^3)^3 (3^3)^4 (2 \cdot 3)^5 \right) = (2^9 \cdot 3^{12} \cdot 2^5 \cdot 3^5) = (2^{9+5} \cdot 3^{12+5}) \\ (36)^2 \cdot 9^4 \cdot (18)^2 = (2^2 \cdot 3^2)(3^2)^4 (2 \cdot 3^2)^2 = (2^4 \cdot 3^4 \cdot 3^8 \cdot 2^2 \cdot 3^4) = (2^{4+2} \cdot 3^{4+8+4}) = (2^6 \cdot 3^{16}) = 2^{14-6} \cdot 3^{17-16} = 2^8 \cdot 3^1$$

On comparing,

$$\alpha = 8, \beta = 1. \text{ And } \alpha + \beta = 9$$

Hence **9 Answer**

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72. The smallest number among

$\sqrt[3]{4}, \sqrt[5]{7}, \sqrt[4]{5}, \sqrt{3}$ is

A $\sqrt[3]{4}$

B $\sqrt[5]{7}$

C $\sqrt[4]{5}$

D $\sqrt{3}$

Answer: B

Explanation:

Solution :

we need to find the LCM of roots' power i.e. LCM of 3,5,4,2 i.e. 60

$$\sqrt[3]{4} = 4^{\frac{1}{3}} = 4^{\left(\frac{20}{60}\right)} = (4^{20})^{\frac{1}{60}}$$

$$\sqrt[5]{7} = 7^{\frac{1}{5}} = 7^{\left(\frac{12}{60}\right)} = (7^{12})^{\frac{1}{60}}$$

$$\sqrt[4]{5} = 5^{\frac{1}{4}} = 5^{\left(\frac{15}{60}\right)} = (5^{15})^{\frac{1}{60}}$$

$$\sqrt{3} = 3^{\frac{1}{2}} = 3^{\left(\frac{30}{60}\right)} = (3^{30})^{\frac{1}{60}}$$

As all the external powers are same, we just need to arrange the internal powers in ascending order to find the lowest one.

On careful observation, and also relying on the fact that 7 has the lowest power. $\sqrt[5]{7}$ is the lowest one. **Answer.**

73. If $(a + b) : (b + c) : (c + a) = 2 : 3 : 4$ and $a + b + c = 9$, then the value of $c =$

A 3

B 1

C 9

D 5

Answer: D

Explanation:

Solution:

$$(a + b) : (b + c) : (c + a) = 2 : 3 : 4$$

Let the common factor of the ratio be x

So,

$$a+b = 2x$$

$$b+c = 3x$$

$$c+a = 4x$$

Adding all the 3 equations

$$2(a+b+c) = 9x$$

A/q

$$a + b + c = 9$$

$$\text{So, } 2.9 = 9x$$

$$\Rightarrow x=2$$

$$\text{So } a+b = 2.2 = 4$$

$$c = (a+b+c) - (a+b) = 9 - 4 = 5 \text{ Answer}$$

74. **A circle and a square have the same area. Then the ratio of the side of the square to the radius of the circle is**

A $\pi : 1$

B $1 : \pi$

C $\sqrt{\pi} : 1$

D $1 : \sqrt{\pi}$

Answer: C

Explanation:

Let circle's radius be r and square's side be a

$$\text{Circle's area} = \pi r^2$$

$$\text{Square's Area} = a^2$$

As both are same ,

$$\pi r^2 = a^2$$

$$\frac{a^2}{r^2} = \pi$$

$$\frac{a}{r} = \sqrt{\pi}$$

$$= \sqrt{\pi} : 1 \text{ Answer}$$

75. $\sqrt{6 + 2\sqrt{2} + 2\sqrt{3} + 2\sqrt{6}} =$

A $1 + \sqrt{2} + \sqrt{6}$

B $1 + \sqrt{2} + \sqrt{3}$

C $\sqrt{2} + \sqrt{3} + 2$

D $2 + \sqrt{3} + \sqrt{6}$

Answer: B

Explanation:

(Hint: taking the cue from the options we should get the term as a square of 3 added terms.)

Now

$$\sqrt{6 + 2\sqrt{2} + 2\sqrt{3} + 2\sqrt{6}} =$$

$$= \sqrt{((1 + 2 + 3) + 2.1.\sqrt{2} + 2.1.\sqrt{3} + 2.\sqrt{2}.\sqrt{3})}$$

$$\text{As } \sqrt{(a + b + c)^2} = \sqrt{a^2 + b^2 + c^2 + 2.a.b + 2.b.c + 2.c.a}$$

On comparison,

$$a = 1, b = \sqrt{2}, c = \sqrt{3}$$

Hence,

$$\sqrt{6 + 2\sqrt{2} + 2\sqrt{3} + 2\sqrt{6}} = 1 + \sqrt{2} + \sqrt{3} \text{ Answer}$$

76. $\left(\frac{\sqrt{5}+\sqrt{7}}{\sqrt{5}-\sqrt{7}} + \frac{\sqrt{5}-\sqrt{7}}{\sqrt{5}+\sqrt{7}} \right)^2 =$

A 196

B 28

C $4\sqrt{35}$

D 144

Answer: D

Explanation:

$$\left(\frac{\sqrt{5}+\sqrt{7}}{\sqrt{5}-\sqrt{7}} + \frac{\sqrt{5}-\sqrt{7}}{\sqrt{5}+\sqrt{7}} \right)^2 = \left\{ \frac{((\sqrt{5}+\sqrt{7})^2 - (\sqrt{5}-\sqrt{7})^2)}{(\sqrt{5}+\sqrt{7})(\sqrt{5}-\sqrt{7})} \right\}^2$$

$$= \left(\frac{((5+7+2.\sqrt{5}.\sqrt{7}) + (5+7-2.\sqrt{5}.\sqrt{7}))}{5-7} \right)^2 = \left(\frac{(12+12)}{-2} \right)^2 = (-12)^2 = 144 \text{ Answer}$$

77. The number that is exactly divisible by 11 among the following is

- A** 9999999
- B** 88888
- C** 1873410
- D** 2345432

Answer: C

Explanation:

Solution'

Concept used: Divisibility rule of 11: The difference of sum of digits at even and odd places should be 0 or multiple of 11

Lets take each no. one by one

	Sum of digits at odd place	Sum of digits at even place	Difference
9999999	$9+9+9+9=36$	$9+9+9=27$	$36-27=9$
88888	$8+8+8=24$	$8+8=16$	$24-16=8$
1873410	$0+4+7+1=12$	$1+3+8=12$	$12-12=0$
2345432	$2+4+4+2=12$	$3+5+3=11$	$12-11=1$

So the required no. is **1873410** Answer

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78. The remainder when $73 \times 79 \times 81$ is divided by 11 is

- A** 1
- B** 5
- C** 7
- D** 9

Answer: A

Explanation:

Principle used to solve : $\text{Rem}(A \times B) = \text{Rem.}(A) \times \text{Rem.}(B)$

A/q

When divided by 11

$$\text{Rem.}(73 \times 79 \times 81) = \text{Rem.}(73) \times \text{Rem.}(79) \times \text{Rem.}(81)$$

$$\text{Rem.}(73) = 7$$

$$\text{Rem.}(79) = 2$$

$$\text{Rem.}(81) = 4$$

$$\text{So Rem.}(73 \times 79 \times 81) = \text{Rem}(7 \times 2 \times 4) = \text{Rem}(56) = 1$$

Hence **Remainder =1** Answer

79. The greatest 4-digit number which is exactly divisible by each of the numbers 18, 24 and 36 is

- A** 1008
- B** 9999
- C** 9972
- D** 9936

Answer: D

Explanation:

Solution

$$18 = 6.3$$

$$24 = 6.4$$

$$36 = 6.6$$

$$\text{LCM} = 6.4.3 = 72$$

So the no. should be divisible by 72. Out of the given options, the Highest such no. before 9999 is 9936.

80. If the l.c.m. of the positive integers a and b is 60 and $a^2.b^2 = 32400$, then the g.c.d. of a and b is

- A** 30
- B** 18
- C** 3
- D** 9

Answer: C

Explanation:

Solution

Concept Used: Product of two numbers = (L.C.M of numbers) x (G.C.D of numbers)

Here

$$a^2.b^2 = 32400$$

$$\text{Hence } a.b = \sqrt{32400} = 180 = \text{Product of Numbers}$$

$$\text{Given L.C.M.} = 60$$

$$\text{So } axb = (\text{l.c.m}) \times (\text{g.c.d})$$

$$180 = 60 \times \text{g.c.d}$$

$$\text{g.c.d} = 3 \text{ Answer}$$

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81. If $|x - 6| = 5$ and $|3y - 12| = 6$ then the maximum value of $\frac{x}{y} =$

- A** 6

B 2

C $\frac{11}{6}$

D $\frac{11}{2}$

Answer: D

Explanation:

Solution

$$|x - 6| = 5 \quad |x-6|=5$$

if $x-6 > 0$; $x > 6$

so the mod function will become, $x-6 = 5$

$x=11$**x max.**

If $x-6 < 0$; $x < 6$

so the mod function will become, $-(x-6) = 5$

$$6 - x = 5$$

$x=6-5=1$ **x min**

Similarly for y

$$|3y - 12| = 6 \quad |3y-12|=6$$

If $3y-12 < 0$, $y < 4$

so the mod function will become, $-(3y-12) = 6$

$$12 - 3y = 6 ; y = 2 \text{ } \mathbf{y \text{ min.}}$$

If $3y-12 > 0$, $y > 4$

so the mod function will become, $(3y-12) = 6$

$$3y = 12+6 = 18$$

$y = 6$ **y max**

For, maximum value of $y = \frac{x}{y}$ Numerator (x) should be maximum and denominator (y) should be minimum.

$$\frac{x(\text{max})}{y(\text{min})} = \frac{11}{2} \quad \mathbf{\text{Answer}}$$

82. If 3 is added to the denominator of a rational number then that number becomes $\frac{1}{3}$ and if 4 is added to numerator of the same rational number, then it becomes $\frac{3}{4}$ then that rational number is

A $\frac{4}{9}$

B $\frac{3}{7}$

C $\frac{5}{12}$

D $\frac{7}{12}$

Answer: C

Explanation:

Solution:

Let the rational No. be $\frac{A}{B}$

A/Q

$$\frac{A}{B+3} = \frac{1}{3} \Rightarrow 3A = B+3$$

$$3A = B+3 \Rightarrow A = \frac{1}{3}(B+3) \dots(i)$$

$$\frac{A+4}{B} = \frac{3}{4} \Rightarrow 4A+16 = 3B \dots(ii)$$

$$4A+16 = 3B$$

$$\frac{4}{3}(B+3) + 16 = 3B$$

$$\frac{4B}{3} + 4 + 16 = 3B$$

$$20 = 3B - \frac{4B}{3}$$

$$20 = \frac{5B}{3}$$

$$B=12$$

Putting in Equation (i)

$$A = 5$$

Hence required rational No. is $\frac{5}{12}$ **Answer**

83. The difference between the biggest and smallest fractions among $\frac{1}{2}, \frac{5}{6}, \frac{7}{9}, \frac{4}{5}$ is

A $\frac{1}{3}$

B $\frac{5}{18}$

C $\frac{3}{10}$

D $\frac{1}{4}$

Answer: A

Explanation:

Solution

$$\frac{1}{2}, \frac{5}{6}, \frac{7}{9}, \frac{4}{5}$$

$$\text{LCM of denominators} = \text{LCM}(2,6,9,5) = 90$$

Converting all the fractions to common base

$$\frac{1}{2} = \frac{45}{90}$$

$$\frac{5}{6} = \frac{75}{90}$$

$$\frac{7}{9} = \frac{70}{90}$$

$$\frac{4}{5} = \frac{72}{90}$$

$$\frac{4}{5} = \frac{4.18}{5.18} = \frac{72}{90}$$

As denominators are same, so fraction with biggest numerator is biggest and with smallest numerator is smallest.

$$\text{Biggest fraction} = \frac{75}{90}$$

$$\text{Smallest fraction} = \frac{45}{90}$$

$$\text{Difference} = \frac{75}{90} - \frac{45}{90} = \frac{30}{90} = \frac{1}{3} \quad \text{Answer}$$

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84. The ascending order of the rational numbers $\frac{13}{14}, \frac{11}{12}, \frac{15}{16}, \frac{8}{11}$ is

A $\frac{8}{11} < \frac{13}{14} < \frac{15}{16} < \frac{11}{12}$

B $\frac{8}{11} < \frac{13}{14} < \frac{11}{12} < \frac{15}{16}$

C $\frac{8}{11} < \frac{11}{12} < \frac{13}{14} < \frac{15}{16}$

D $\frac{13}{14} < \frac{11}{12} < \frac{15}{16} < \frac{8}{11}$

Answer: C

Explanation:

Solution:

$$\frac{13}{14}, \frac{11}{12}, \frac{15}{16}, \frac{8}{11}$$

We can proceed with approximation;

$$\frac{13}{14} = 0.92..$$

$$\frac{11}{12} = 0.91...$$

$$\frac{15}{16} = 0.93...$$

$$\frac{8}{11} = 0.72....$$

So, Ascending Order will be 0.72, 0.91, 0.92, 0.93

$$\text{i.e. } \frac{8}{11}, \frac{11}{12}, \frac{13}{14}, \frac{15}{16}$$

85. If 7% of 5% of 8% of x is 105, then x =

A 375

B 3750

C 37500

D 375000

Answer: D

Explanation:

Solution:

A/q

7% of 5% of 8% of x is 105

$$\frac{7}{100} \times \frac{5}{100} \times \frac{8}{100} x = 105$$

$$7.5.8.x = 105000 \times 1000$$

$$280x = 105000000$$

$$x = 375000 \text{ Answer}$$

86. If the length and breadth of a rectangle are increased by 20% each, then the area of the rectangle increases by x%, then x =

A 20

B 120

C 44

D 144

Answer: C

Explanation:

Solution:

Let Original dimensions be l and b.

$$\text{Area} = l \times b$$

New Dimension be L and B

$$\% \text{ Increase in } l = 20$$

$$L = (100+20)\% \text{ Of } l = 1.2l$$

$$\% \text{ Increase in } b = 20$$

$$B = (100+20)\% \text{ Of } b = 1.2b$$

$$\text{New Area} = L \times B = (1.2l)(1.2b) = 1.44 lb$$

$$\begin{aligned} \% \text{ change increase in area} &= \frac{\text{Difference In Area}}{\text{Old Area}} \times 100 \\ &= \frac{1.44lb - lb}{lb} \times 100 = 44\% \text{ Answer} \end{aligned}$$

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87. An article when sold at a gain of 5% yields rupees 15 more than when it is sold at a loss of 5%. Then the cost price of that article in rupees is

A 75

- B** 100
C 150
D 200

Answer: C

Explanation:

Solution:

Gain % = 5

Loss % = 5

Difference in profit percentages = $5 - (-5) = 10\%$

This 10% (of CP) corresponds to monetary value of Rs 15

1% of CP corresponds to Rs. $15/10 = \text{Rs. } 1.5$

100% of CP = $\text{Rs. } 1.5 \times 100 = \text{Rs. } 150$ **Answer**

88. By selling 15 items, a seller recovers the cost price of 20 items. Then his profit percent is

- A** 25
B $33\frac{1}{3}$
C 20
D $35\frac{1}{5}$

Answer: B

Explanation:

Solution

A/q

20 C.P. = 15 S.P.

15 C.P. + 5 C.P. = 15 S.P.

5 C.P. = 15 S.P. - 15 C.P.

$C.P = \frac{15}{5} (S.P. - C.P.) = 3 (S.P. - C.P.)$

$C.P = 3 (S.P. - C.P.)$

$\frac{1}{3} = \frac{(S.P. - C.P.)}{C.P} = \text{Fractional Profit}$

Profit percentage = Fractional Profit $\times 100 = \frac{100}{3} \%$

$= 33\frac{1}{3} \%$ **Answer**

89. A, B, C entered into a business with a total capital of ₹ 50,000/-. A invested ₹ 4,000/- more than B and B invested ₹ 5,000/- more than C. If there is a total profit of ₹ 35,000/- then A's share in the profit, in rupees is

- A** 8,400
- B** 11,900
- C** 14,700
- D** 15,000

Answer: C

Explanation:

Solution

Let C's investment be C

So

$$B = C + 5000$$

$$A = B + 4000 = (C + 5000) + 4000 = C + 9000 = A$$

$$\text{So } A + B + C = 50000$$

$$C + 9000 + C + 5000 + C = 50000$$

$$14000 + 3C = 50000$$

$$3C = 36000$$

$$C = 12000$$

$$\text{Hence } A = C + 9000 = 21000$$

$$B = C + 5000 = 17000$$

$$\text{Ratio of } A:B:C = 21:17:12$$

$$\text{Profit} = 35000$$

$$\text{As share} = \frac{21}{21+17+12} \times 35000 = \frac{21}{50} \times 35000 = 21 \times 700$$

$$= 14700 \text{ Answer}$$

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90. A, B and C are running a business by investing the capitals in the ratio 2 : 4 : 5. If C's share in the profit is ₹ 2,200/-, then A's share in the profit, in rupees, is

- A** 400
- B** 800
- C** 840
- D** 880

Answer: D

Explanation:

Solution:

A, B and C's investments are in the ratio 2 : 4 : 5.

So profits will be shared in the same ratio.

For every Rs.(2+4+5)=11 profit, C gets Rs 5 and A gets Rs 2

So if C gets Rs 5...A gets Rs 2

if C gets Rs 1...A gets Rs $\frac{2}{5}$

if C gets Rs 2200...A gets Rs $\frac{2}{5} \times 2200 = \text{Rs } 4400$

= Rs.880 Answer

91. Two pipes A and B can fill a tank independently in 20 and 30 minutes respectively. Both the pipes are opened simultaneously and after 6 minutes tap B is closed. Then the total time taken, in minutes, to fill the tank, is (assume that the tank is empty initially)

A 10

B 15

C 16

D 12

Answer: C

Explanation:

Solution:

Let total work be 60 units [LCM of 20,30]

A does 60 units in 20 minutes. Hence in 1 minute A does $(60/20)\text{units} = 3 \text{ units}$.

B does 60 units in 30 minutes. Hence in 1 minute B does $(60/30)\text{units} = 2 \text{ units}$.

Total work in 1 minute = 5 units

Now, for the given scenario,

First,

A and B work for 6 minutes together @ 5 units/min

Work done = $(5 \times 6) \text{ units} = 30$

Work remaining = Total units of work - work done = $60 - 30 = 30 \text{ UNITS}$.

Then,

A work alone @ 3 units/min.

Time taken = $(\text{Work remaining} / \text{speed}) = 30/3 \text{ mins.} = 10 \text{ minutes}$

Total time = $6 + 10 = 16 \text{ minutes}$

92. Tap A can fill an empty tank in one hour, while a drain pipe B can empty that full tank in 6 hours. If both A and B are opened at the same instant, the total time taken to fill that empty tank, in hours, is

A $\frac{5}{6}$

B $\frac{3}{2}$

C $\frac{2}{3}$

D $\frac{6}{5}$

Answer: D

Explanation:

Solution:

Work done by A in 1 hour = 1 unit

Work done by B in 1 hour = $-\frac{1}{6}$ unit

Total work done by A and B in 1 hr. = $1 - \frac{1}{6} = \frac{5}{6}$

Time taken to fill the tank = $\frac{1}{\left(\frac{5}{6}\right)} = \frac{6}{5}$ hours
 = $\frac{6}{5}$ hours **Answer**

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93. A train travels a distance of 60 km in 40 minutes. If its velocity is decreased by 15 km/hour, then the time required by that train to travel the same distance, in minutes, is

A 45

B 60

C 54

D 48

Answer: D

Explanation:

Solution

Distance = 60km

Time = 40 minutes = $\frac{40}{60}$ hour = $\frac{2}{3}$ Hour

Speed = $\frac{\text{Distance}}{\text{Time}}$

Speed = $\frac{60}{\left(\frac{2}{3}\right)} = \frac{(60)(3)}{2} = 90$ kmph

Reduced speed = $90 - 15 = 75$ kmph

Distance = 60 km

Speed = $\frac{\text{Distance}}{\text{Speed}}$

Time = $\frac{60}{75} = \frac{4}{5}$ hr. = 48 minutes.

94. A person travels from A to B at a speed of 75 km/hour in a car and returns from B to A by reducing his speed by 15 km/hour. If the total time taken is 3 hours, then the distance between A and B, in kilometers, is

- A** 100
- B** 80
- C** 120
- D** 160

Answer: A

Explanation:

Solution:

Let, The distance between A-B be d .

While going from A to B Speed = 75 kmph

Time = $\frac{\text{Distance}}{\text{Speed}} = \frac{d}{75}$ hours

While going from B to A Speed = 75 kmph

Time = $\frac{\text{Distance}}{\text{Speed}} = \frac{d}{75 - 15} = \frac{d}{60}$ hours

Total time = 3 hours

$\Rightarrow \frac{d}{75} + \frac{d}{60} = 3$

$\frac{d}{75} + \frac{d}{60} = 3$

$\frac{4d + 5d}{300} = 3$

$\frac{9d}{300} = 3$

$d = 100$ km **Answer**

95. Two persons A, B can together complete a piece of work in 12 days, B and C together complete the same work in 8 days. If A, B, C work together, then that work can be finished in 6 days. Then the number of days B alone can finish the same work is

- A** 16
- B** 18
- C** 24
- D** 30

Answer: C

Explanation:

Solution

A, B can together complete the work in 12 days

Hence efficiency of (A+B) = $\left(\frac{100}{12}\right)\% = 8\frac{1}{3}\%$

B and C together complete the same work in 8 days.

Hence efficiency of (B+C) = $\left(\frac{100}{8}\right)\% = 12\frac{1}{2}\%$

A, B, C work together, then that work can be finished in 6 days

Hence efficiency of $(A+B+C) = \left(\frac{100}{6}\right)\% = 16\frac{2}{3}\%$

Efficiencies

$$A+B = 8\frac{1}{3}\% \quad \dots\dots(i)$$

$$B+C = 12\frac{1}{2}\% \quad \dots\dots(ii)$$

$$A+B+C = 16\frac{2}{3}\% \quad \dots\dots(iii)$$

Solving for B,

iii - ii

$$(A+B+C) - (B+C) = A = 16\frac{2}{3}\% - 12\frac{1}{2}\% = 4\frac{1}{6}\% = \frac{25}{6}\%$$

Putting this in (i)

$$\frac{25}{6}\% + B = 8\frac{1}{3}\%$$

$$B = 8\frac{1}{3}\% - \frac{25}{6}\% = \frac{25}{6}\%$$

$$\text{Time taken by B} = \left(\frac{1}{\text{efficiency}}\right) = \frac{100}{\frac{25}{6}} = 24$$

24 days Answer

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96. A, B, C can individually complete a work in 20 days, 15 days, 12 days respectively. B and C start the work and they worked for 3 days and left. Then the number of days required by A to finish the remaining work is

A 11

B 14

C 12

D 9

Answer: A

Explanation:

Solution

A, B, C can individually complete a work in 20 days, 15 days, 12 days respectively

$$\text{Efficiency of } (A + B + C) = \left(\frac{1}{20} + \frac{1}{15} + \frac{1}{12}\right) = \left(\frac{3+4+5}{60}\right) = \frac{12}{60}$$

$$\text{Efficiency of only } (B + C) = \left(\frac{1}{15} + \frac{1}{12}\right) = \left(\frac{4+5}{60}\right) = \frac{9}{60}$$

$$\text{So efficiency of A} = \frac{12}{60} - \frac{9}{60} = \frac{3}{60}$$

While working, all 3 work for 3 days.

$$\text{So work completed in 3 days} = 3 \times \frac{12}{60} = \frac{36}{60}$$

$$\text{Work Remaining} = 1 - \frac{36}{60} = \frac{24}{60}$$

$$\text{If A works alone @ } \frac{3}{60} \text{ work per day.} = \frac{\left(\frac{24}{60}\right)}{\left(\frac{3}{60}\right)}$$

{60}\right)) = 8 days.

97. If the length of the hypotenuse of right angled Iscoceles triangle is 12 cm , then the area of the triangle in sqaure centimeters is

- A 72
B 36
C 144
D 24

Answer: B

Explanation:

Solution

Let Perpendicular be **p**, base be **b** and Hypotenuse be **h**

As it is isosceles right angle triangle,

so $P=B$

According to Pythagoras Theorem

$$h = \sqrt{p^2 + b^2}$$

$$12 = \sqrt{b^2 + b^2}$$

$$12 = \sqrt{2b^2}$$

$$12 = \sqrt{2} \cdot b$$

$$b = \frac{12}{\sqrt{2}}$$

$$\text{Area} = \frac{1}{2} \times \text{product of sides} \times \sin \text{ of included angle}$$

$$= \frac{1}{2} \times b \times b \times \sin 90^\circ$$

$$= \frac{1}{2} \times \left(\frac{12}{\sqrt{2}} \right)^2 \times 1$$

$$= \frac{1}{2} \times 72 \times 1$$

=36 Answer

98. If the length of the diagonals of a rhombus are 18 cm and 24 cm then the area of a rhombus in square centimeters is

- A 108
B 225
C 432
D 216

Answer: D

Explanation:

Solution

Area of Diagonal = $\frac{1}{2} \left(\text{product of diagonals} \right) = \frac{1}{2} (18 \times 24) = 216 \text{ cm}^2$

AP SI Free Solved Previous Papers PDF

99. A cylindrical bar of height 1 metre and base radius 0.75 metres is melted and cast as a sphere and cast as a sphere. Then the diameter of a sphere in meters is

- A 2
- B 1.5
- C 0.75
- D 0.50

Answer: B

Explanation:

Solution:

Volume of Cylindrical bar = Volume of cast sphere

$$\pi r^2 h = \frac{4}{3} \pi R^3$$

$$\pi (0.75)^2 \cdot 1 = \frac{4}{3} \pi R^3$$

$$\left(\frac{3}{4} \right)^2 = \frac{4}{3} R^3$$

$$\left(\frac{3}{4} \right)^2 \cdot \frac{3}{4} = R^3$$

$$\left(\frac{3}{4} \right)^3 = R^3$$

$$R = \frac{3}{4} \text{ m}$$

$$D = 2R = \frac{3}{2} = 1.5 \text{ m}$$

00. Two boxes have square bases. The sides of these bases are in the ratio 2 : 1. If the heights of the boxes are in the ratio 1 : 2 respectively, then the ratio of the volumes of the boxes in that order is

- A 1 : 1
- B 1 : 2
- C 3 : 1
- D 2 : 1

Answer: D

Explanation:

Solution:

Bases (b) are in the ratio 2:1

Height (h) are in the ratio 1:2

As, volume's formula is defined as $b^2 h$

So Volume ratio $\Rightarrow \frac{\frac{1}{3}\pi(2)^2h}{\frac{1}{3}\pi(1)^2h} = \frac{4}{1} = 4:1$

Answer = **2:1**

01. The radius of a cone is 3 times the radius of a cylinder and their heights are same. Then the ratio of their volumes in that order is

A 9 : 1

B 1 : 3

C 3 : 1

D 1 : 9

Answer: C

Explanation:

Solution:

The radius of the cone is 3 times the radius of the cylinder and their heights are same.

Volume of cone = $\frac{1}{3}\pi R^2h$

Volume of cylinder = πr^2h

Given, $R = 3r$

So, Volume of cone = $\frac{1}{3}\pi (3r)^2h$

Ratio of volumes = $\frac{\frac{1}{3}\pi (3r)^2h}{\pi r^2h} = 3:1$ Answer

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02. A window is in the shape of a rectangle surmounted by a semicircle. If the length of the rectangle is 6 feet and the radius of the semicircle is 2 feet, then the area of the window, in square feet, is

A $18 + 4\pi$

B $24 + 4\pi$

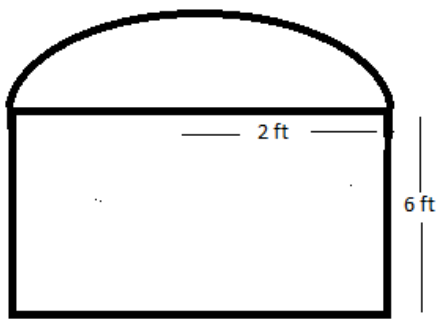
C $24 + 2\pi$

D $18 + 2\pi$

Answer: C

Explanation:

Solution:



Area of the window = Area of rectangular part + area of semicircular part

$$= l \cdot b + \frac{\pi r^2}{2} \quad \dots (\text{Radius of semicircle} = 0.5b)$$

$$= 6 \times (2 \cdot 2) + \frac{\pi (2)^2}{2} = 24 + 2\pi$$

$$= 24 + 2\pi$$

03. 11 cubic meters of steel is cast in to cylindrical bars of diameters 10 cm and the length 1.4 metres. Then the number of such bars that can be cast with the given metal is (Take $\pi = \frac{22}{7}$)

- A 10
- B 100
- C 1000
- D 10000

Answer: C

Explanation:

Solution

Volume of given steel = 11 m^3

$$\text{Volume of the bar} = \pi \left(\frac{0.1}{2} \right)^2 \cdot 1.4 \text{ m}^3 = \frac{22}{7} \cdot \frac{0.01}{4} \cdot 1.4 \text{ m}^3 = 11 \cdot \frac{0.01}{2} \cdot 0.2 = 11 \cdot 0.01 \cdot 0.1 = 0.011$$

$$\text{No. of bars, } n = \frac{\text{Volume of steel}}{\text{Volume of a bar}} =$$

$$\frac{11}{11 \times 10^{-3}} = 10^3 = 1000 \text{ Answer}$$

04. The difference between the largest three digit natural number x satisfying $x \equiv 8 \pmod{8}$ and the smallest three digit natural number y satisfying $y \equiv 2 \pmod{5}$ is

- A 895
- B 0
- C 1
- D 995

Answer: A

Explanation:

Solution

y will be in the form of $5x+2$

SSC CHSL Previous Question papers (download pdf)

05. If $3^{48} = x \pmod{10}$ and if $0 \leq x \leq 9$ then $x =$

- A 1
- B 3
- C 5
- D 9

Answer: A

Explanation:

For,

$$3^{48} = x \pmod{10}$$

We need to find remainder when 3^{48} is divided by 10.

3 has a cyclicity of 4. So as 48 is a multiple of 4, so last digit will be 1.

Hence remainder on division by 10 = 1

1 Answer

06. If p and q are two statements, then which one of the following is a tautology?

- A $p \vee (\sim p)$
- B $p \vee (\sim q)$
- C $(\sim p) \wedge q$
- D $p \wedge q$

Answer: A

Explanation:

Solution

p	$\sim p$	$p \vee (\sim p)$
0	1	1
1	0	1

Hence

Option A Answer

07. Which of the following is equivalent to the statement $P \rightarrow Q$?

- A $P \wedge Q$
- B $P \vee Q$

C $\sim P \wedge Q$

D $\sim Q \wedge P$

Answer: C

Explanation:

None of the options

SSC CHSL Free Mock Test

08. Set A has 5 elements and set B has 10 elements. Then the possible maximum number of elements in $A \cup B$ is

A 5

B 10

C 15

D 50

Answer: C

Explanation:

Solution

For maximum no. of elements, they have to be disjoint

$$n(A \cup B) = n(A) + n(B)$$

$$= 5 + 10 = 15 \text{ Answer}$$

09. Let A be non empty set and P(A) be the power set of (A). Let R be a relation defined on P(A) by $X R Y \iff X \subseteq Y \text{ for all } X, Y \in P(A)$. Then R is not

A an a symmetric relation

B a transitive relation

C a symmetric relation

D a reflexive relation

Answer: C

10. If a function $f: R \rightarrow R$ is defined by $f(x) = x^2 + 1$, then $f^{-1}(26) =$

A the empty set

B $\left\{ \sqrt{26}, -\sqrt{26} \right\}$

C $\{5, -5\}$

D $\{0\}$

Answer: C

Explanation:

Solution

$$\text{If } f(x) = x^2 + 1, \text{ then } f^{-1}(26) =$$

$$\text{Let } f(x) = y$$

$$y = x^2 + 1, \text{ then } f^{-1}(26) =$$

$$x = \sqrt{y-1}$$

putting $y = 26$

$$f^{-1}(26) = \sqrt{26-1} = \sqrt{25} = 5$$

Range = $(-5, +5)$ Answer

SSC CGL Free Mock Test

11. If the line passes through the points $(3, -5)$ and $(-2, 7)$, then x is the intercept is

A $\frac{-5}{11}$

B $\frac{12}{11}$

C $\frac{11}{12}$

D $\frac{11}{5}$

Answer: C

Explanation:

Solution

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Putting the values,

$$y_2 = 7, x_2 = -2$$

$$y_1 = -5, x_1 = 3$$

$$y - (-5) = \frac{7 - (-5)}{-2 - 3} (x - 3)$$

$$y + 5 = \frac{12}{-5} (x - 3)$$

$$-5(y + 5) = 12(x - 3) \Rightarrow -5y - 25 = 12x - 36$$

on rearranging,

$$12x + 5y = 11$$

For x intercept

putting $y = 0$

$$x = \frac{11}{12} \text{ Answer}$$

12. The length of the intercept made by the line which passes through the points (2, 5) and (-4, 8) between the coordinate axis

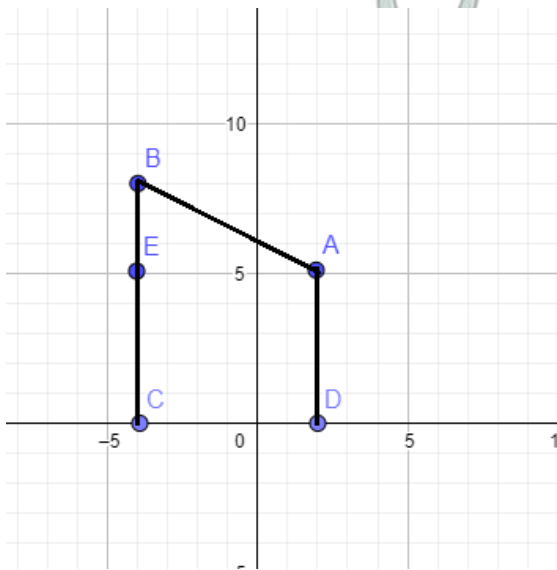
- A $12\sqrt{5}$
- B $12\sqrt{3}$
- C $6\sqrt{5}$
- D $6\sqrt{3}$

Answer: C

Explanation:

Solution

A/q



Here A = (2, 5) and B = (-4, 8)

intercept length = CD = AE.

Also, AEB is a right angle triangle right angled at E.

So ACC. to pythagoras theorem

$$\sqrt{BA^2 - BE^2} = AE = CD$$

=

13. If θ lies in the first quadrant and $5 \tan \theta = 4$, then $\frac{5 \sin \theta - 3 \cos \theta}{\sin \theta + 2 \cos \theta} =$

- A $\frac{1}{14}$
- B $\frac{2}{14}$
- C $\frac{3}{14}$
- D $\frac{5}{14}$

Answer: D

Explanation:

Solution

$$\text{Given } 5 \tan \theta = 4$$

$$\text{So } \tan \theta = (4/5)$$

$$P = 4$$

$$B = 5$$

$$H = \sqrt{4^2 + 5^2} = \sqrt{41}$$

$$\sin \theta = (4/\sqrt{41})$$

$$\cos \theta = (5/\sqrt{41})$$

So

$$\frac{5 \sin \theta - 3 \cos \theta}{\sin \theta + 2 \cos \theta}$$

$$= \{5(4/\sqrt{41}) - 3(5/\sqrt{41})\} / \{(4/\sqrt{41}) + 2(5/\sqrt{41})\}$$

$$= \{20\sqrt{41} - 15\sqrt{41}\} / \{4\sqrt{41} + 10\sqrt{41}\}$$

$$= \frac{5}{14} \text{ Answer}$$

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14. If $\theta + \phi = \frac{\pi}{4}$, then $(1 + \tan \theta)(1 + \tan \phi) =$

A -1**B** 0**C** 1**D** 2**Answer: D****Explanation:**

Solution

$$\text{If } \theta + \phi = \frac{\pi}{4}, \text{ then } (1 + \tan \theta)(1 + \tan \phi) =$$

$$\text{We can assume, } \theta = 0$$

$$\text{So } \phi = \frac{\pi}{4}$$

then putting values in,

$$(1 + \tan \theta)(1 + \tan \phi) =$$

$$= (1 + 0)(1 + \tan \frac{\pi}{4}) = (1 + 0)(1 + 1) = 1 \times 2 = 2$$

$$= 2$$

15. $(\cos^2 15^\circ - \cos^2 75^\circ)(\sin 65^\circ \cdot \cos 35^\circ - \cos 65^\circ \cdot \sin 35^\circ)$
 $=$

A $\frac{1}{2}$

B $\frac{\sqrt{3}}{2}$

C $\sqrt{3}$

D $\frac{\sqrt{3}}{4}$

Answer: D

Explanation:

Solution:

$$\begin{aligned} & (\cos^2 15^\circ - \cos^2 75^\circ) (\sin 65^\circ \cdot \cos 35^\circ - \cos 65^\circ \cdot \sin 35^\circ) = \\ & \left(\cos 15^\circ + \cos 75^\circ \right) \left(\cos 15^\circ - \cos 75^\circ \right) \left(\sin 65^\circ \cos 35^\circ - \cos 65^\circ \sin 35^\circ \right) \\ & \left(2 \cos \left(\frac{90}{2} \right) \right) \left(\cos \left(\frac{60}{2} \right) \right) \left(2 \sin \left(\frac{90}{2} \right) \right) \left(\sin \left(\frac{60}{2} \right) \right) \left(\sin 65^\circ \cos 35^\circ - \cos 65^\circ \sin 35^\circ \right) \\ & \left(2 \cos 45^\circ \right) \left(\cos 30^\circ \right) \left(2 \sin 45^\circ \right) \left(\sin 30^\circ \right) \left(\sin 65^\circ \cos 35^\circ - \cos 65^\circ \sin 35^\circ \right) \\ & \left(2 \cdot \frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{3}} \right) \left(2 \cdot \frac{1}{\sqrt{2}} \right) \left(\frac{1}{2} \right) \left(\sin 30^\circ \right) \\ & \left(2 \cdot \frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{3}} \right) \left(2 \cdot \frac{1}{\sqrt{2}} \right) \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) \end{aligned}$$

On solving

$\frac{\sqrt{3}}{4}$ **Answer**

16. Two poles of height 15 metres and 30 metres stand upright on a play ground .If the feet of the poles are 36 meters apart , then the distance between their tops are

A 35 metres

B 37 meters

C 39 meters

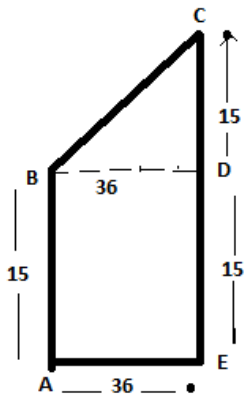
D 41 meters

Answer: C

Explanation:

Solution

A/q



$$AB = DE = 15\text{m}$$

$$\text{So } CD = CE - AB = 15\text{m}$$

Also given $AE = 36\text{m}$. It implies $BD = 36$.

Hence in right angle triangle BCD.

$$\text{Base} = 36\text{m}$$

$$\text{Height} = 15\text{ m}$$

So according to Pythagoras theorem Hypotenuse (or distance between their tops) will be **39m. Answer**

SSC CGL PREVIOUS PAPERS (download pdf)

17. If the polynomial $ax^4 + bx^3 + 3x^2 - 4x - 4$ is divisible by $(x^2 - 1)$ then $(a, b) =$

A (4,2)

B (1,4)

C (-1,4)

D (2,-4)

Answer: B

Explanation:

Solutions The Polynomial is divisible by $(x^2 - 1)$, so it will have $(x^2 - 1)$ as one of its factors.

$$\text{Also, } (x^2 - 1) = (x+1)(x-1)$$

So both +1 and -1 will be the roots of the polynomial.

Hence $f(1)$ and $f(-1)$ will be equal to 0.

$$f(1) = a(1) + b(1) + 3(1) - 4 - 4 = a + b + 3 - 4 - 4 = a + b - 5 = 0 \dots (i)$$

$$f(-1) = a(-1) + b(-1) + 3(-1) + 4 - 4 = -a - b + 3 = 0 \dots (ii)$$

On solving (i) and (ii)

$$a = 1 \text{ and } b = 4. \text{ Hence Answer (1,4)}$$

18. If $(x - \alpha)$ and $(x - \beta)$ are the two factors of the polynomial $f(x) = ax^2 + bx + c$, then the quadratic polynomial whose factors are $(x - \frac{1}{\alpha})$ and $(x - \frac{1}{\beta})$ is

- A** $cx^2 - bx + a$
- B** $cx^2 + bx + a$
- C** $cx^2 + bx - a$
- D** $cx^2 - bx - a$

Answer: B

Explanation:

Solution:

If factors are $(x - \alpha)$ and $(x - \beta)$ then roots of the expression will be α, β .

Product of roots = $\alpha \times \beta = (c/a)$ Eqn (i)

Sum of roots = $\alpha + \beta = -(b/a)$ Eqn (ii)

In the new case,

Let polynomial be $Mx^2 + Nx + O$

when factors are $(x - \frac{1}{\alpha})$ and $(x - \frac{1}{\beta})$

Roots will be $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

So product of roots = $O/M = (1/\alpha) \times (1/\beta) = (1/\alpha\beta) = a/c$ from equation (i)

Sum of roots = $-N/M = (1/\alpha) + (1/\beta) = \{(\alpha + \beta) / \alpha\beta\} = (-b/a) / (c/a) = -b/c$from equation (ii)

So product of roots = $a/c = O/M$

Sum of roots = $-b/c = -N/M$

On comparison, the polynomial $Mx^2 + Nx + O$ turns out to be

$$cx^2 + bx + a$$

19. If $63 \times 65 \times 67 \times 69$ is divided by 12 then the remainder obtained is

- A** 9
- B** 7
- C** 5
- D** 3

Answer: A

Explanation:

Solution:

Principle used to solve : $\text{Rem}(A \times B) = \text{Rem.}(A) \times \text{Rem.}(B)$

A/q

When divided by 12

$\text{Rem.}(63 \times 65 \times 67 \times 69) = \text{Rem.}(63) \times \text{Rem.}(65) \times \text{Rem.}(67) \times \text{Rem.}(69)$

The common multiple of 12 smaller than all the above nos. is 60. So taking 60 as reference.

Rem. (63) $\times$ Rem. (65) $\times$ Rem. (67) $\times$ Rem. (69) = Rem. (3 $\times$ 5 $\times$ 7 $\times$ 9) = Rem. (945)

945 **leaves a remainder of 9** when divided by 12, as 936 is the closest multiple of 12 before 945.

Hence **Answer 9**

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20. A polynomial in x leaves remainders 8, 4 when divided by $(x + 2)$ and $(x - 2)$ respectively. If the same polynomial is divided by $x^2 - 4$ then the remainder is

- A 12
- B $x + 10$
- C $6 - x$
- D 32

Answer: C

Explanation:

A polynomial in x leaves remainders 8, 4 when divided by $(x + 2)$ and $(x - 2)$ respectively. If the same polynomial is divided by $x^2 - 4$ then the remainder is

Let fn. be $f(x)$.

As $f(x)$ leaves remainder, 8 when divided by $(x+2)$

Therefore, $f(-2) = 8$

When $f(x)$ is divided by $(x-2)$, remainder = 4

Therefore, $f(2) = 4$

Now, polynomial is divided by $x^2 - 4$ or $(x+2)(x-2)$.

the remainder will be linear. in the form of $ax+b$.

$$f(x) = q(x)(x+2)(x-2) + r(x)$$

When, $x=2$,

$$\Rightarrow f(2) = 0 \cdot (x-2)q(2) + r(2) = 4 \Rightarrow f(1) = 0 \cdot (x-2)q(1) + r(1) = 4$$

$$r(2) = 2a + b = 4 \dots (i)$$

When, $x=-2$

$$\Rightarrow f(-2) = 0 \cdot (x+2)q(-2) + r(-2) = 8 \Rightarrow f(-2) = 0 \cdot (x+2)q(-2) + r(2) = 8$$

$$r(-2) = (-2a) + b = 8 \dots (ii)$$

Solving above two equations, we get remainder $r(x) = 6 - x$ **Answer**

21. If $x + y = 12$, $y + z = 20$ and $z + x = 18$ then $x - y + z =$

- A 25
- B 18

C 11

D 7

Answer: C

Explanation:

given that

$$x + y = 12 \quad \text{equation 1}$$

$$y + z = 20 \quad \text{equation 2}$$

$$z + x = 18 \quad \text{equation 3}$$

on adding equation 1 + 2 + 3 we get

$$x + y + z = 25 \quad \text{equation 4} \quad \text{now perform operation eq4 - eq1, eq4 - eq2, and eq4 - eq3}$$

we get

$x = 5, y = 7$ and $z = 13$ then put the value in eq following

$$x - y + z = 5 - 7 + 13 = 11 \quad \text{Answer}$$

22. A Person has a certain number of two rupee coins and some five rupee coins. If the total number of coins is 16 and the value of coins is ₹50, then the number by which the number of two rupee coins exceeds number of five rupee coins is

A 5

B 6

C 4

D 2

Answer: C

Explanation:

let he have x number of 2 Rs coin and y number of 5 Rs coin

then we can say that

$$2x + 5y = 50 \quad \text{equation 1}$$

$$x + y = 16 \quad \text{equation 2} \quad \text{on solving equation 1 and 2 we get}$$

$$x = 10 \quad \text{number of 2 Rs coin}$$

$$y = 6 \quad \text{number of 5 Rs coin}$$

$$\text{then by question } x - y = 10 - 6 = 4 \quad \text{Answer}$$

SBI Clerk Free Mock Test (Latest Pattern)

23. If the first and sixth terms of geometric progressions are $\frac{2}{3}$ and 162 respectively then the 8^{th} term of the progression is

- A 324
- B 729
- C 1084
- D 1458

Answer: D

Explanation:

let the first term is a and common ratio is r then

$$T_1 = a \times r^{1-1} = a = \frac{2}{3} \quad \text{equation 1}$$

$$T_6 = a \times r^{6-1} = a \times r^5 = 162 \quad \text{equation 2}$$

equation 1 \ equation 2

$$\frac{2}{3} \times 162 = 1 \times r^5$$

$$243 = r^5$$

$$3 = r$$

$$r = 3$$

put the value of a and r in equation following equation

$$T_8 = a \times r^{8-1} = \frac{2}{3} \times 3^7$$

$$T_8 = \frac{2}{3} \times 243 \times 3 \times 3 \times 3$$

$$T_8 = 1458 \quad \text{Answer}$$

24. Three numbers are in an arithmetic progression. their sum is 3 The third number is greater than first number by 16 ,then the product of those three number is

- A -63
- B -36
- C 12
- D 24

Answer: A

Explanation:

Let x-d, x and x+d be the three numbers.

Then in question given that

$$x - d + x + d + x = 3 \text{ or } x = 1 \text{ and also given that}$$

$$x + d - x + d = 16 \text{ or } d = 8$$

then we get the sequence = -7, 1 and 9 so then multiplication on all term

we get -63 **Answer**

25. The term independent of x in the expansion of $\left(3x^3 - \frac{4}{x}\right)^8$ is

- A $2^{14} \times 3 \times 7^2$
- B $2^{12} \times 3^3 \times 7^2$
- C $2^{14} \times 3^2 \times 7$
- D $2^{12} \times 3^4 \times 7$

Answer: C

Explanation:

Let $(r+1)^{\text{th}}$ term be the independent of x which is given by binomial theorem

$$T_{r+1} = {}^8C_r \cdot (3x^3)^{8-r} \cdot \left(-\frac{4}{x}\right)^r$$

here we have $a = 3x^3$ and $b = -\frac{4}{x}$

put the value we get

$$T_{r+1} = {}^8C_r \cdot (3x^3)^{8-r} \cdot (-4/x)^r$$

$$T_{r+1} = {}^8C_r \cdot (3)^{8-r} \cdot (x^3)^{8-r} \cdot (-4)^r \cdot (x)^{-r}$$

$$T_{r+1} = {}^8C_r \cdot (3)^{8-r} \cdot (x^{24-3r}) \cdot (-4)^r \cdot (x)^{-r}$$

since the term is independent of x , we have

$$24 - 4r = 0$$

$$r = 6$$

so the $(r+1) = 7$ term is independent so we can say that

$$T_7 = {}^8C_6 \cdot (3)^{8-6} \cdot (-4)^2 \cdot (6-1)$$

$$T_7 = 8 \cdot {}^8C_2 \cdot 3^2 \cdot (-4)^2 \cdot 5$$

$$T_7 = 2^{14} \cdot 7 \cdot 3^2 \cdot 5 \quad \text{Answer}$$

SBI Clerk Previous Papers (Download PDF)

26. If P is the sum of odd terms and Q is the sum of even terms in the expansion of $(x + y)^n$ then $P^2 - Q^2 =$

- A $(x^2 - y^2)^n$
- B $(x - y)^{2n}$
- C $(x + y)^{2n}$
- D $(x^2 - y^2)^{2n}$

Answer: A

Explanation:

we know that the expansion of sum of even term and sum of odd term are respective given below

$$(a + x)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} x + {}^nC_2 a^{n-2} x^2 + \dots + {}^nC_n x^n$$

$$+ \dots + n {}^nC_n x^n$$

$${}^n(a-x)^n = {}^nC_0 a^n - {}^nC_1 a^{n-1} x + {}^nC_2 a^{n-2} x^2 - \dots + {}^nC_n (-1)^n x^n$$

adding both we get

$${}^n(a+x)^n + {}^n(a-x)^n = 2({}^nC_0 a^n + {}^nC_2 a^{n-2} x^2 + \dots + n {}^nC_n x^n)$$

hence

$$P = ({}^n(a+x)^n + {}^n(a-x)^n)^{1/2} \quad \text{equation 1}$$

$$Q = ({}^n(a+x)^n - {}^n(a-x)^n)^{1/2} \quad \text{equation 2}$$

there fore PQ we get

$$PQ = ({}^n(a+x)^n + {}^n(a-x)^n)^{1/2} \times ({}^n(a+x)^n - {}^n(a-x)^n)^{1/2} = ({}^n(a+x)^{2n} + {}^n(a-x)^{2n})^{1/4}$$

$$\text{or } 4PQ = ({}^n(a+x)^{2n} + {}^n(a-x)^{2n})$$

$${}^n P^2 - {}^n Q^2 = (({}^n(a+x)^n + {}^n(a-x)^n)^{1/2})^2 - (({}^n(a+x)^n - {}^n(a-x)^n)^{1/2})^2$$

$${}^n P^2 - {}^n Q^2 = {}^n(a+x)^n {}^n(a-x)^n$$

$${}^n P^2 - {}^n Q^2 = (a^2 - x^2)^n \quad \text{Answer}$$

27. If A is a 3 x 3, square matrix and if $A \text{ Adj } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then $\det(2A) =$

A 4

B 32

C 8

D 64

Answer: B

Explanation:

given that $A \text{ Adj } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

means that

$$A \text{ Adj } A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = |A| = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A| = 4$$

$$\det(2 \text{ Adj } A) = |A|^{(n-1)}, \text{ where } n \text{ is number of rows in a matrix}$$

$$= 2 \times 4^{(3-1)}$$

$$= 2 \times 16$$

$$= 32 \quad \text{Answer}$$

$$28. \lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 7x + 10} =$$

- A** $\frac{7}{3}$
- B** 0
- C** $\frac{-3}{5}$
- D** does not exist

Answer: A

Explanation:

given that

$$\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 7x + 10}$$

we can write this as

$$\lim_{x \rightarrow 5} \frac{x^2 - 5x + 2x - 10}{x^2 - 5x - 2x + 10} = \lim_{x \rightarrow 5} \frac{x(x-5) + 2(x-5)}{x(x-5) - 2(x-5)} =$$

$$\lim_{x \rightarrow 5} \frac{(x-5)(x+2)}{(x-5)(x-2)} = \lim_{x \rightarrow 5} \frac{(x+2)}{(x-2)} =$$

put the value of x we get

$\frac{7}{3}$ **Answer**

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29. If $x = t - \frac{1}{t}$, $y = t + \frac{1}{t}$ where t is a parameter then $\frac{dy}{dx} =$

- A** $\frac{x}{y}$
- B** $\frac{y}{x}$
- C** $\frac{-x}{y}$
- D** $\frac{-y}{x}$

Answer: A

Explanation:

given that

$$x = t - \frac{1}{t}, y = t + \frac{1}{t}$$

then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{1 + \frac{1}{t^2}}{1 - \frac{1}{t^2}}$$

$$\frac{dy}{dt} = 1 + \frac{1}{t^2} \text{ and } \frac{dx}{dt} = 1 - \frac{1}{t^2}$$

$$\frac{dy}{dx} = \frac{1 + \frac{1}{t^2}}{1 - \frac{1}{t^2}} = \frac{t^2 + 1}{t^2 - 1} = \frac{x+y}{x-y} \text{ Answer}$$

30. If A is a square matrix and A^T denotes transpose then the matrix AA^T is always

- A** a symmetric matrix

- B** a skew symmetric matrix
- C** a orthogonal matrix
- D** a null matrix

Answer: A

Explanation:

we know that the

$$A.A^T = A^T.A$$

so when we solve this always get symmetric matrix we found

31. When a piece of wire is bent in the form of equilateral triangle then the area of the triangle is $121\sqrt{3} \text{ cm}^2$. If the same piece of wire is bent in the form circle, then the area of the circle in square centimeters is (Take $\pi = \frac{22}{7}$)

- A** 258.5
- B** 346.5
- C** 364.5
- D** 385.5

Answer: B

Explanation:

we know that the area of the eqitriangle = $a^2 \times \sqrt{3} \div 4 = 121\sqrt{3}$

$$a = 22$$

then the length of the wire = perimeter of the triangle

$$= 3a = 3 \times 22 = 66$$

then the radius of the circle = $2\pi r = 2 \times \frac{22}{7} \times r = 66$

$$r = 21 \div 2$$

then area of the circle = $\pi \times r^2 = \frac{22}{7} \times \frac{21^2}{2}$

then area of the circle = 346.5 cm square **Answer**

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32. The sum of all the internal angles of a regular hexagon, in radians, is

- A** 2π
- B** 4π
- C** 3π
- D** 5π

Answer: B

Explanation:

we know that the sum of all the internal angles of a regular hexagon, in radians, is $= 720 = 4\pi$

Answer

33. If two circles of radii, 7 cm and 10 cm respectively touch each other internally, then the distance between their centres, in centimeters, is

A 17

B 3

C -3

D 5

Answer: B

Explanation:

If circles are touching internally the distance between their centers will be = radius of bigger circle - radius of smaller circle

$$= 10 - 7$$

$$= 3 \text{ cm}$$

Answer

note -

If circles are touching externally the distance between their centers will be = radius of bigger circle + radius of smaller circle

34. If the mid points of the sides AB and AC of a triangle ABC are (4, -2), (-8, 14) respectively, then the length of the side BC, in units, is

A 20

B 30

C 40

D 60

Answer: C

Explanation:

let the mid points of the sides AB and AC of the triangle is D(4,-2) and F(-8,14)

which is parallel to the BC and $BC = 2DF$

then DF Length is $DF = \sqrt{(-8-4)^2 + (14+2)^2} = 20$

then $BC = 2 \times 20 = 40$ **Answer**

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35. The distance between the lines $\frac{x}{3} + \frac{y}{4} = 1$ and $8x + 6y = 5$ in units, is

- A $\frac{9}{5}$
- B $\frac{9}{10}$
- C $\frac{19}{5}$
- D $\frac{19}{10}$

Answer: D

Explanation:

we know the formula of distance between two line

$$\frac{(A_1x + B_1y + C) \div \sqrt{(A_1^2 + B_1^2)}}{\pm \frac{(A_2x + B_2y + C) \div \sqrt{(A_2^2 + B_2^2)}}{\sqrt{(A_1^2 + B_1^2) + (A_2^2 + B_2^2)}}}$$

then we have here $A_1 = 1, B_1 = 1, C_1 = -1$ and $A_2 = 8, B_2 = 6, C_2 = -5$

$$A_1 = 1, B_1 = 1, C_1 = -1 \text{ and } A_2 = 8, B_2 = 6, C_2 = -5$$

then

$$\frac{(A_1x + B_1y + C) \div \sqrt{(A_1^2 + B_1^2)}}{\pm \frac{(A_2x + B_2y + C) \div \sqrt{(A_2^2 + B_2^2)}}{\sqrt{(A_1^2 + B_1^2) + (A_2^2 + B_2^2)}}}$$

$$\frac{(1x + 1y - 1) \div \sqrt{1^2 + 1^2}}{\pm \frac{(8x + 6y - 5) \div \sqrt{8^2 + 6^2}}{\sqrt{1^2 + 1^2 + 8^2 + 6^2}}}$$

$$1 + 9 \div 10 = 10 \div 19 \text{ Answer}$$

36. The mean of 6 observations is calculated as 24 and later it was noted that one of the observations is wrongly taken as 24 instead of -24 then the correct mean is

- A 40
- B 32
- C 24
- D 16

Answer: D

Explanation:

when we calculated mean we ascending data

here if we take -24 instead of the 24 so it comes in first place as it is smallest value and affect first 3 value of 6 observation

$$\text{so } (24 - (-24)) \div 3 = 16 \text{ new mean Answer}$$

37. The median of 15 observations is 32. Each of the observations greater than the median are increased by 8 and each of the observations less than the median are decreased by 6. Then the median of new data is

- A 46
- B 18

C 32

D 40

Answer: C

Explanation:

Since 8 is less than 32 and 32 is more than 6, so the middle value remains unchanged and hence the median of new data observations taken together will be 32.

Answer

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38. The mode of the following distribution: 19, 13, 16, 13, 19, 24, 13, 9, 24, 13, 26 is

A 26

B 17

C 13

D 9

Answer: C

Explanation:

Mode is the observation that occurs most frequently in the data

here 13 is repeated 3 times so 13 is our answer

39. If the standard deviation of the first n natural numbers is t then $12t^2 + 1 =$

A $n^2 - 1$

B n^2

C $n^2 + 1$

D $n^2 + 2$

Answer: B

Explanation:

The standard deviation of n natural numbers = t

Therefore, but we know that

First n natural numbers are $1, 2, 3, \dots, n$.

Mean = $(1 + 2 + 3 + \dots + n) \div n$

$= [n(n+1)/2] / n$

$(n+1)/2 \dots (i)$

And, $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)(2n+1)/6$

Or, $(1^2+2^2+3^2+\dots+n^2)/n = \{(n+1)(2n+1)/6\}$... (ii)

Variance of first n natural numbers $\sigma^2 = \{(1^2+2^2+3^2+\dots+n^2)/n\} - (\text{Mean})^2$

$$= \{(n+1)(2n+1)/6\} - \{(n+1)^2/4\}$$

$$= (n+1)/12 \{4n+2-3(n+1)\}$$

$$= (n+1)(n-1)/12$$

$$\sigma^2 = (n^2-1)/12$$

and we have the equation given

$$12t^2 + 1 = \sigma^2$$

$t^2 = -1/12$ put the of t we get

n^2 **Answer**

40. If the standard deviation of a variable x is 12 then the variance of the variable $2x + 17$ is

A 24

B 288

C 576

D 41

Answer: A

Explanation:

Let say

$$Y = 2X + 17$$

Taking variance of both sides:

$$V(Y) = V(2X) + V(17)$$

$$V(Y) = 2 V(X) + 0$$

$$V(Y) = 2 * 12$$

$$V(Y) = 24$$

Answer

$$V(aX) = a^2 * V(X) \text{ .. where a is a constant}$$

$$V(a) = 0 \text{ .. where a is a constant}$$

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41. If 3 coins are tossed, then the probability of getting at least 2 heads is

A $\frac{1}{2}$

B $\frac{1}{8}$

C $\frac{1}{4}$

D $\frac{3}{8}$

Answer: A

Explanation:

when the 3 coin are tossed the total number of cases are = $2^3 = 8$

and at least two heads are there in the following = HHT, HTH, THH, HHH = 4

then the probability of getting at least two heads = number of favourable cases / total number of cases

then the probability of getting at least two heads = $\frac{4}{8} = \frac{1}{2}$ **Answer**

42. When two unbiased dice are thrown, the probability that they show up different numbers is

A $\frac{1}{6}$

B $\frac{11}{36}$

C $\frac{5}{6}$

D $\frac{25}{36}$

Answer: C

Explanation:

when the two unbiased dice are thrown the number of total cases are = 36

let the both the dice show the same number then cases are = (1,1), (2,2), (3,3), (4,4), (5,5) and (6,6) = 6

then the probability of getting different number = 1 - probability of getting same number on both dice

then the probability of getting different number = $1 - \frac{6}{36}$

then the probability of getting different number = $1 - \frac{1}{6}$

then the probability of getting different number = $\frac{5}{6}$ **Answer**

43. In a bivariate distribution $X_i, y_i, i = 1, 2, \dots, 8$, if d_i is the deviation between the ranks of X_i and Y_i and $\sum_{i=1}^8 d_i^2 = 21$ then the coefficient of rank correlation between X_i and Y_i is

A 0.25

B 0.50

C 0.65

D 0.75

Answer: D

Explanation:

If $X_1, X_2, \dots, X_n$ are mutually independent normal random variables with means $\mu_1, \mu_2, \dots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, then the linear combination

of ranks X_i and Y_i means have equal value 4 and 4 as $i = 1, 2, 3, 4, \dots, 8$

$$Y = \sum_{i=1}^8 d_i^2 = 21$$

Then, finding the probability that X is greater than Y reduces to a normal probability calculation:

$$P(X > Y) = P(X - Y > 0) = P(Z > 0 - 1/4) = P(Z > -0.75) = P(Z < 0.75) = 0.75 \quad \text{Answer}$$

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44. If 5 boys and 4 girls sit in a row at random then the probability the boys and girls sit alternatively is

- A $\frac{1}{63}$
- B $\frac{1}{126}$
- C $\frac{1}{120}$
- D $\frac{1}{60}$

Answer: B

Explanation:

Given that 5 boys and 4 girls sit in a row at random.

Total number of ways in which 9 persons can be arranged in a row = $9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

= 5 boys can be arranged themselves in $5 \times 4 \times 3 \times 2 \times 1$ ways.

B 1 B 2 B 3 B 4 B

Now,

Then there are 4 positions left(1,2,3,4).

The 4 girls can be seated in (1,2,3,4) = $4 \times 3 \times 2 \times 1$ ways.

The girls can be arranged themselves in $4 \times 3 \times 2 \times 1$ ways.

Therefore, the number of ways both girls and boys can sit alternatively = $5 \times 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1$

Hence,

Required probability =

$$\frac{(5 \times 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1)}{(9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1)}$$

$$= \frac{(120 \times 24)}{(9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1)}$$

$$= 4/504$$

$$= 1/126. \quad \text{Answer}$$

45. When a leap year is selected at random then the probability that it has exactly 52 Sundays is

- A** $\frac{2}{7}$
- B** $\frac{5}{7}$
- C** $\frac{1}{7}$
- D** $\frac{6}{7}$

Answer: B

Explanation:

The question is asking the probability of getting "exactly" 52 Sundays and not at least 52 Sundays.

There are 52 weeks in a year, so there will be 52 Sundays, no doubts about it. However, $52 \times 7 = 364$ days

so that still leaves a day for a normal year, and two days for a leap year.

Now, let us look at the combinations of the two days possible.

(Sunday, Monday), (Monday, Tuesday), (Tuesday, Wednesday), (Wednesday, Thursday), (Thursday, Friday), (Friday, Saturday), (Saturday, Sunday)

Now, to get exactly 52 Sundays, none of the last two days should be a Sunday. As you can see from the combinations listed above, that is possible in 5 out of 7 cases. So the Probability would be $\frac{5}{7}$. **Answer**

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Communication Ability

Instructions [146 - 162]

Choose the correct answer :

46. **A situation in which goods or shares are plentiful and buyers can keep prices down is called**

- A** a buy out
- B** a buyer's market
- C** a business proposition
- D** a business deal

Answer: B

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47. **A difference in wages between industries or between categories of Employees in the same industry referred to as**

- A** differential
- B** differential equation

C differential calculus

D differentiate

Answer: A

48. **A large corportion formed by the merging of separate firms is known as**

A Congregation

B Conglomeration

C Conclave

D Consul

Answer: B

49. **Celebraties who endorsed a product reffered to as**

A Brand Managers

B Brand Ambassadors

C Super Models

D Glamour Kings

Answer: B

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50. **Stocks ,Shares ,Bonds are collectively known as**

A Cumulative Preferance shares

B Dividends

C Securities

D Equalities

Answer: C

51. **A computer language that is used to create websites is known as**

A X-files

B XML

C XL

D X-Certificate

Answer: B

52. connection of networks that can be joined together is possible through

A Intranet

B Extranet

C Virtual private network

D Internet

Answer: D

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53. ISDN stands for

A Integrated Services Digital Network

B Integrated Subscribers Digital Network

C International Services Developers Network

D Integrated Services Developers Network

Answer: A

54. ELD stands for

A Electric Lamination Display

B Energy Luminiscent Display

C Electro Luminiscent Display

D Energy Lubrication Display

Answer: C

55. The connection between a computer and another piece of equipment is

A parallel processing

B parallel
bar

C parallel port

D parallelism

Answer: C

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56. **Choose the Correct Answer.**

A: "How is your business ?"

B: "I'm broke!"

B's answer indicates that

- A** he has taken a break
- B** he is heart-broken
- C** he has suffered heavy loss
- D** he has broken a record

Answer: C

57. **A: "you should not have spoken so harshly to him. you know that the faulty was yours "**

B: "I know. I'll have to eat humble pie "

- A** easy to forget
- B** atome
- C** apologise
- D** forgive himself

Answer: C

58. **A: " I have two tickets for the movie tommorrow .Will you come with me ? "**

B: "could you invite some one else ? movies give me a headache ."

- A** ill
- B** busy
- C** not A's friend
- D** not interested in movies

Answer: D

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59. **"I dont think his jokes ever come off"**

The speakers implies that the jokes were

- A** unsuccessful

- B** brilliant
- C** uninspiring
- D** unwelcome

Answer: A

60. The passive form of the sentence "They are not going to play the match today " is

- A** The match is not going to be played today
- B** The match was not played today
- C** The match had not been played today
- D** The match can not be played today

Answer: A

61. **A:** "I'm sorry I couldn't come to your party as I was out of town."

B: "It's alright. Don't worry about it".

In this conversation, the speaker "B" is

- A** indifferent
- B** disinterested
- C** understanding
- D** critical

Answer: C

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62. "When it comes to grammar, she's really on the ball."

The speaker implies that

- A** she knows alot about grammar
- B** she's bad at grammar
- C** she's lazy
- D** she doesn't like grammar

Answer: A

Instructions [163 - 170]

Fill in the blanks with the appropriate phrase/verb/preposition.

63. A: "Are you going to the bank today?"

B: "No. I've already to the bank."

- A been
- B gone
- C went
- D made

Answer: A

64. I hope the enquiry will truth

- A get behind
- B get at
- C get by
- D get away with

Answer: B

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65. The workers their resignation when their demands were not met by the managing committee

- A gave in
- B gave up
- C called off
- D took in

Answer: A

66. Here is the watch you asked

- A for
- B with
- C by
- D on

Answer: A

67. The loan will be repaid two months

- A** by
- B** from
- C** for
- D** in

Answer: D

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68. It raining for a while , but now its raining again.

- A** finished
- B** stopped
- C** is
- D** goes

Answer: B

69. They are the whole family for Christmas.

- A** bringing over
- B** bringing about
- C** bringing up
- D** taking up

Answer: A

70. A subordinate has to abide the decisions ofhis superiors.

- A** with
- B** in
- C** by
- D** Up on

Answer: C

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Instructions [171 - 175]

Read the following passage and answer questions :

In times of social fragmentation, vulgarity often becomes a way of life. To be shocking becomes more important than to be civil or creative or truly original. Given the degree of vulgarity in our society, cynicism seems almost irresistible. But cynicism represents a secession from society, a dissolution of the bonds between people and families and communities, an indifference to the fate of anything, beyond the self. Cynicism is deadly. It bites everything it can reach like a dog with a foot caught in a trap. And then it devours itself. It drains us of the will to improve; it diminishes our public spirit; it saps our inventiveness; it withers our souls. Cynics often see themselves as merely being world-weary. There is no new thing under the sun, the cynics say. They claim their weariness is wisdom. But it is usually mere posturing. Their weariness seems to be most effective when they consider the aspirations of those beneath them, who have neither power nor influence nor wealth. For these unfortunates, nothing can be done, the cynics declare. Hope is considered an affront to rationality: the notion that the individual has a responsibility for the community is considered a dangerous radicalism. Ultimately, however, the life of a cynic is lonely and self-destructive. It is our human nature to make connections with other human beings. The gift of sympathy for another is one of the most powerful sentiments we ever feel. If we do not have it, we are not human. Indeed it is so powerful that the cynics who deny it go to war with himself.

71. **What is Considered normal in today's society?**

- A** creativity
- B** courtesy
- C** a lack of good taste
- D** bad habits

Answer: C

72. **What have divisions in the society led to**

- A** sanity
- B** rationality
- C** depression
- D** a belief that people always act from selfish motives.

Answer: D

73. **Why cynicism dangerous**

- A** its leads to weariness
- B** Its self destructive
- C** its kills our creativity
- D** It is regressive

Answer: B

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74. What characterises the behaviour of a cynic ?

- A Affectation
- B Humility
- C Generosity
- D Foolishness

Answer: A

75. What quality makes us human?

- A Genuine concern for others
- B Quest for power
- C Quest for money
- D A radical attitude

Answer: A

Instructions [176 - 180]

Read the following passage and answer questions

Production may not be a dominant ideological leverage in economic decision making, but it cannot be altogether dismissed from economic calculations at whatever level. The difference that globalization has brought about is not that it has made production concerns immaterial, but that the leading factor is now the consumer and production must do what it can to meet the demands of the market. So every now and again, one comes up against serious production problems at the national level which are resolved at the global level by importing skills or machinery. Thus while it is true that consumer tastes are getting more homogenized across the world, production facilities have not kept up. Economic difficulties at the national level arise from this mismatch.

76. What cannot be kept out of economic calculations ?

- A Ideology
- B Decision making
- C Market demand
- D Production

Answer: D

77. What is the most prominent consideration today ?

- A The consumer
- B The economy
- C Production
- D Globalisation

Answer: D

78. Which word does the writer use to describe buyer's tastes ?

- A Immaterial
- B Mismatch
- C Homogenised
- D Ideological

Answer: C

79. What has caused the change in approach to economic decision-making ?

- A Consumer tastes
- B Globalisation
- C Imported machinery
- D Production problems

Answer: D

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80. At the national level, what is the effect of the discrepancy between production and demand ?

- A Importing skills becomes necessary
- B Production problems need to be addressed
- C Demands of the consumer call for urgent steps
- D Economic difficulties arise in the process

Answer: A

Instructions [181 - 185]

Read the following passage and answer questions

The land in Egypt is very fertile when it is irrigated, and water for irrigation has to come mainly from the one

big river which Egypt has, namely, the Nile. Therefore, most of the farming is done near this river. As the Nile nears the sea, it splits into several branches to form a delta. The land here is very flat; ploughing and irrigation are easy and it is, therefore, the most highly cultivated area. Further inland, away from the sea, only the strips of land on either side of the river are cultivated. There are, of course, a few irrigation projects and model farms, but a large proportion of the Egyptian country side remains a dry, sandy desert. In the delta, the main crops are cotton, rice, fruit such as banana, dates, figs and melons, and vegetables. Cattle, buffalo, and goats are also kept for milk and meat. Further inland, however, where the land is not so flat, and water is scarce, the cultivation is less intense; in this part of the country, there is a scarcity of fresh food, such as fruit and vegetables, because there is not enough water to grow it. Sugarcane and wheat are the main crops here. In the huge, barren deserts, there are wandering herdsmen who keep goats and camels. Thus it is clear that vast areas of land are lying unused at present only because of the lack of irrigation. The country can double its production if the irrigation projects multiply.

81. **What is the meaning of 'irrigation' ?**

- A** To build dams
- B** To supply land with water
- C** To construct canals
- D** To cultivate land

Answer: B

82. **What does the Nile form when it reaches the sea ?**

- A** A triangular alluvial tract
- B** An island
- C** A lake
- D** A tributary

Answer: A

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83. **What are cultivated towards the interior of the city ?**

- A** Bananas and dates
- B** Sugarcane and wheat
- C** Cotton and rice
- D** Vegetables and fruits

Answer: B

84. **Who are the 'wandering herdsmen' ?**

- A** Men with no homes
- B** Men who roam with their livestock
- C** Men who tend goats and cattle
- D** Men who live in the desert

Answer: B

85. How can agricultural production be increased in Egypt ?

- A** By cultivating all the land
- B** By introducing new crops
- C** By making water available to uncultivated lands
- D** By growing more fruits

Answer: C

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Instructions [186 - 191]

Choose the correct meaning for the word given:

86. **bulwark**

- A** freedom
- B** protection
- C** shipment
- D** bundle

Answer: B

87. **cower**

- A** desert
- B** timid
- C** hide
- D** run

Answer: C

88. **plebiscite**

- A** statement of loyalty
- B** promoter of monarchy
- C** countrywide vote
- D** political analyst

Answer: C

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89. **bizarre**

- A** breezy
- B** depressing
- C** very strange
- D** bright

Answer: C

90. **Platitude**

- A** common place
- B** plateau
- C** extraordinary
- D** misleading

Answer: A

91. **feign**

- A** faint
- B** fill
- C** feel
- D** pretend

Answer: D

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Instructions [192 - 195]

Fill in the blank choosing the correct word:

92. **Conspirators were satirically called men.**

- A** honourable
- B** wicked
- C** lazy
- D** active

Answer: A

93. **The of information in the organization will be given high priority.**

- A** dissemination
- B** dissipation
- C** distortion
- D** disbursement

Answer: A

94. **Bitter experiences make us though wiser.**

- A** better
- B** fortunate
- C** sadder
- D** obvious

Answer: C

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95. **The of the flowers enhanced the beauty of the garden.**

- A** fragrance
- B** odour
- C** smell
- D** aroma

Answer: A

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